Topics of today's class

Last class:

- Chapter 7
  - Economic growth facts
  - Malthusian model of economic growth

Today's class:

- Chapter 7
  - Solow growth model
  - Growth accounting

Solow Growth Model

- This model is the basis for the modern theory of economic growth.
- A key prediction is that technological progress is necessary for sustained increases in standards of living.

South Korea and the Philippines

- In 1960, South Korea and the Philippines were quite similar
  - Per capita GDP: Korea=$1800, Philippines=$2200
  - Population: Both about 25 million, 1/2 working age
  - Similar sectorial composition (industry, agriculture)
  - College enrollment: Korea=5%, Philippines=13%
- Between 1960 and 2007, macroeconomic performance diverged sharply:
  - Growth: Korea=5.5%, Philippines=1.7%
  - Per capita GDP in 2007: Korea=$24,000, Philippines=$5,000
- Why?

Population Growth

- In the Solow growth model, population is assumed to grow at a constant rate n.

$$N' = (1 + n)N$$

Consumption-Savings Behavior

- Consumers are assumed to save a constant fraction s of their income, consuming the rest.

$$C = (1 - s)Y$$

Representative Firm's Production Function

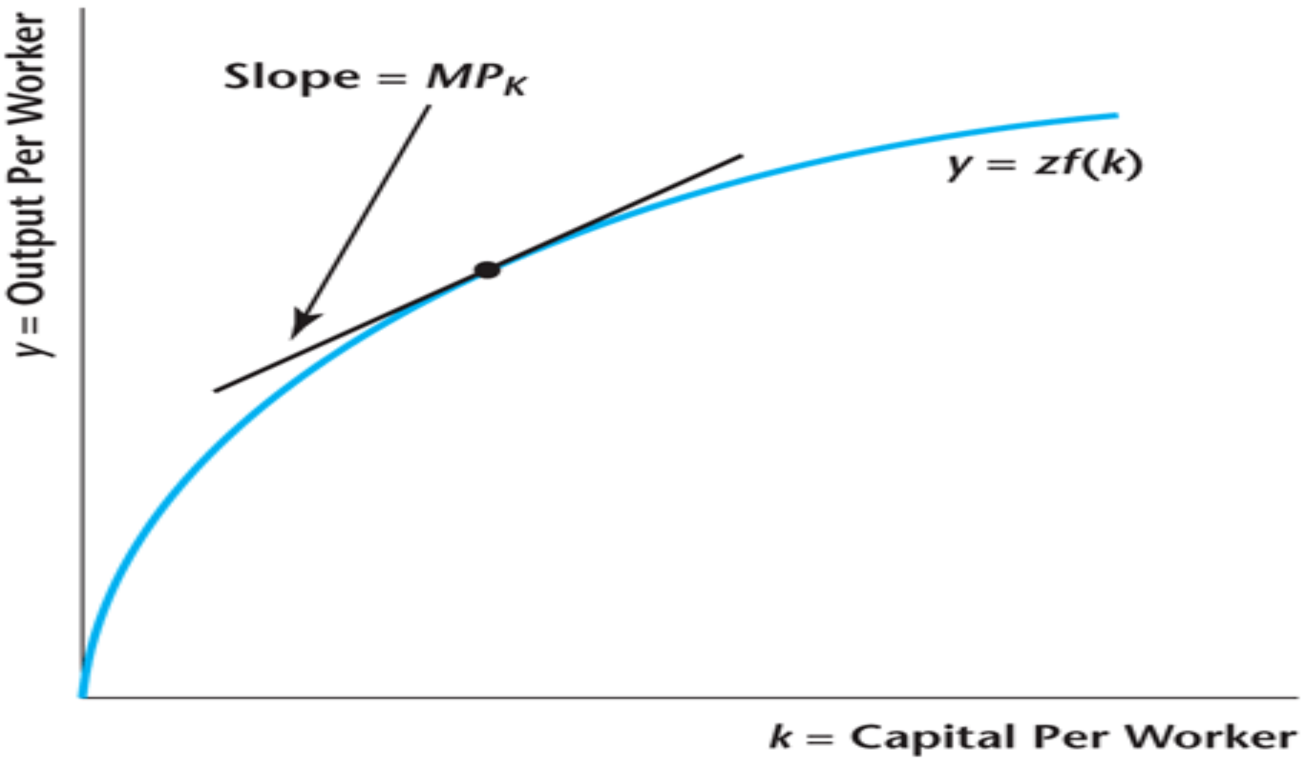
$$Y = zF(K, N),$$

Constant Returns to Scale

- Constant returns to scale implies:

$$Y/N = zF(K/N, 1)$$

Figure 6.12
The Per-Worker Production Function



Evolution of the Capital Stock

- Future capital equals the capital remaining after depreciation, plus current investment.

$$K' = (1 - d)K + I$$

Solving for the Solow model

- Exogenous Variables: s, d, n, z
- Endogenous Variables: Y, K, N, C, I

The income expenditure identity:

$$Y = C + I$$

combined with the household budget constraint:

$$Y = C + S$$

Saving = Investment

$$S = I$$

Solving for the Solow model

- Replacing in the capital accumulation equation

$$K' = K(1 - d) + S$$

$$= K(1 - d) + sY$$

$$= K(1 - d) + szF(K)$$

Rewrite in per-worker form:

$$k'(1 + n) = (1 - d)k + szf(k)$$

This equation governs the dynamics of the Solow Model.

Dynamics of capital per worker

There are two opposing forces driving capital per worker:

- investment
  - consumer saves a constant fraction of her income which is used for investment.
  - ***diminishing returns***: each addition to the capital stock increases production and therefore investment by less and less.
- the growth of the population *n* and depreciation of existing capital *d*.

Figure 6.13
Determination of the Steady State Quantity of Capital per Worker

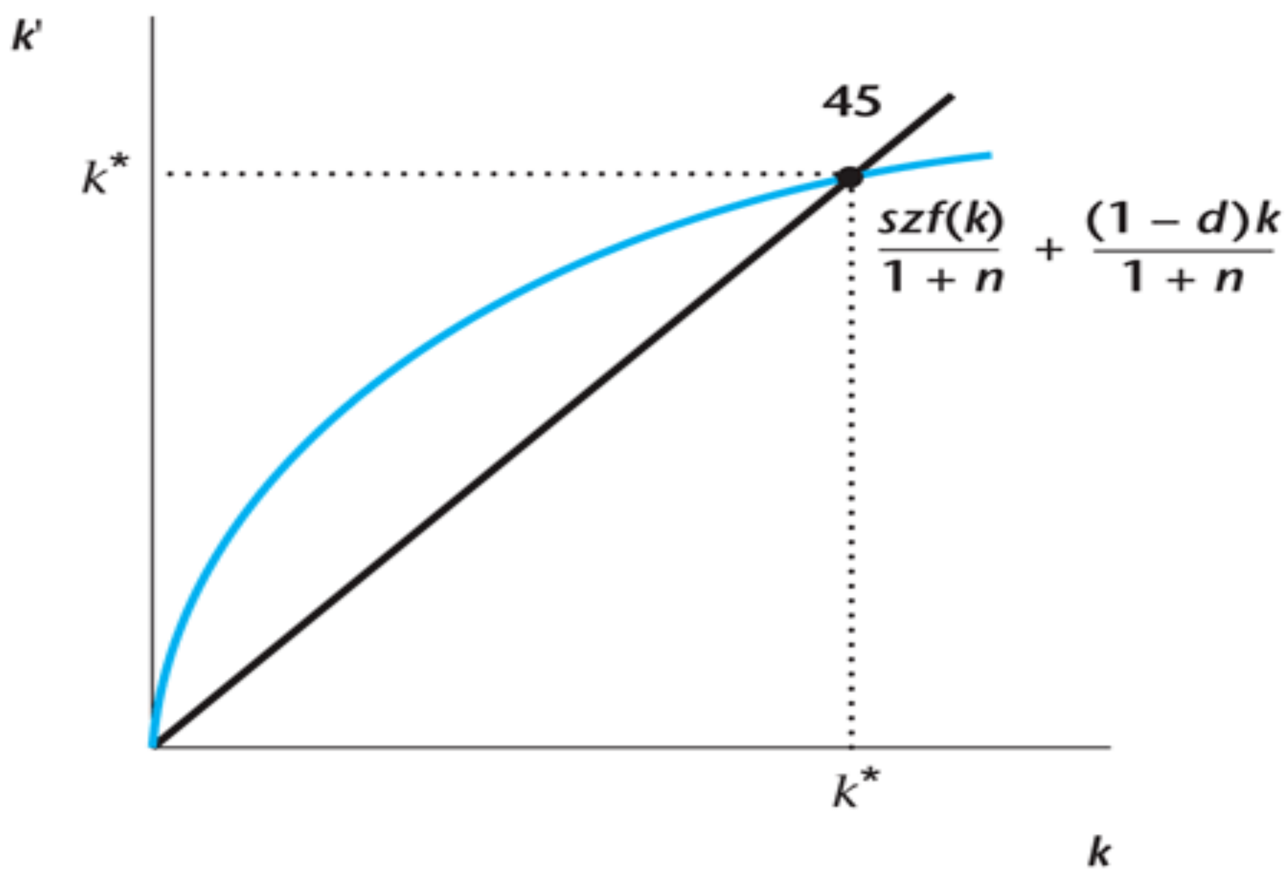
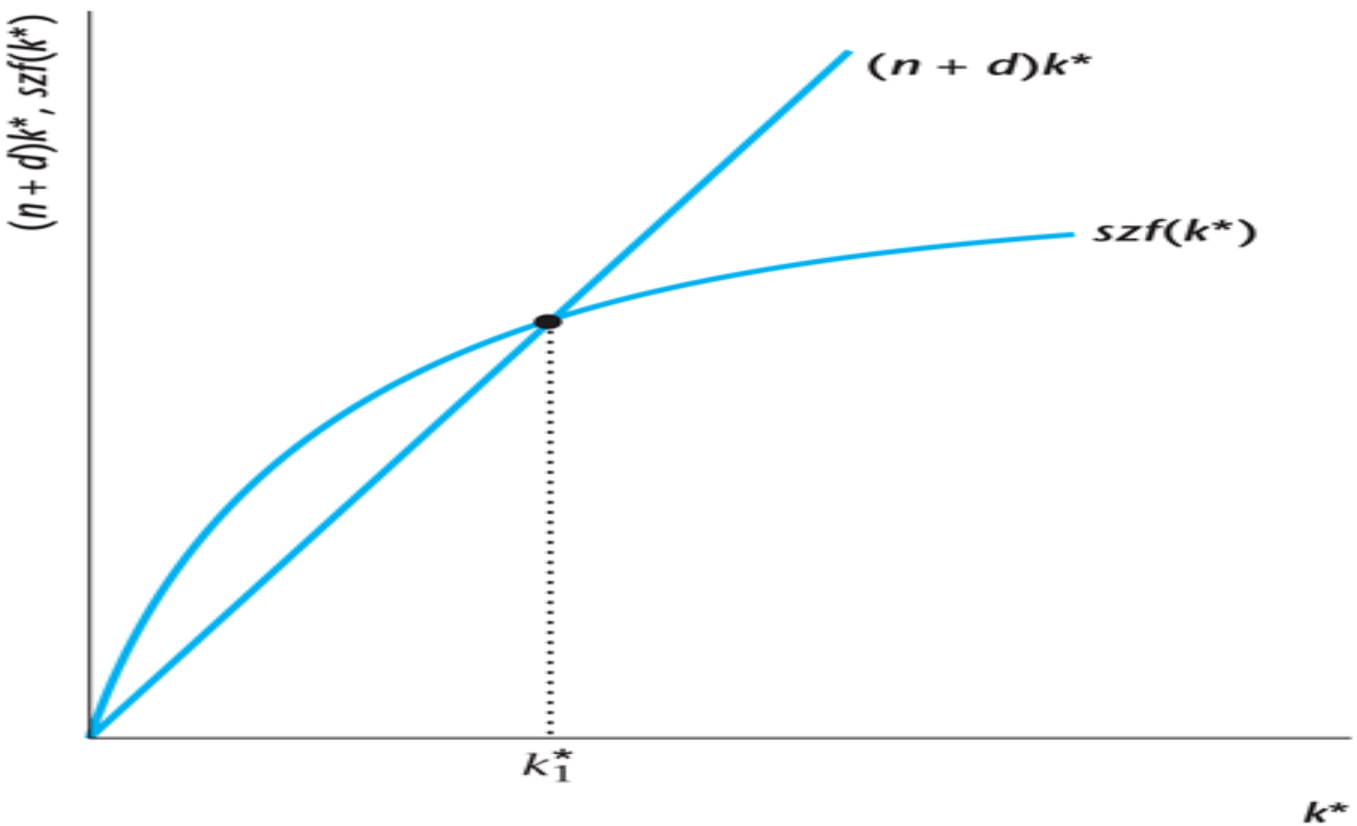


Figure 6.14
Determination of the Steady State Quantity of Capital per Worker



Dynamics of Capital per Worker

Imagine initially low k_0:

- productivity of capital is very high
- the stock of capital increases
- until the economy reaches the steady state k^*

Why does the economy settle down to a steady state?

- Because investment exhibits diminishing returns: as we increase capital, output rises by a smaller and smaller amount.
- But a constant fraction of capital depreciates.

What is the source of Long Term Growth?

- Zero! There is no long-run growth in GDP per capita in Solow!!
  - Diminishing returns to capital is at the heart of why growth eventually ceases in the Solow Model.
- Despite this negative result on long-run growth, the Solow framework is extraordinary helpful: see “Nobel Prize” :-)
- The reason is related to ***Transition Dynamics***.

The Principle of Transition Dynamics

- A key prediction of the Solow framework is

  The Principle of Transition Dynamics*: the farther below its steady state an economy is, the faster it will grow; similarly, the farther above its steady state, the slower it will grow.*

- Differences in growth rates across countries largely reflect how near or far countries are from their steady state.

Example: The United States and China

- United States
  - Growth at a constant rate of 2% per year for more than a century.
  - In some sense, this reflects the fact that the U.S. is close to its steady state.
  - We'll understand this more soon.
- China
  - Rapid growth at more than 8% per year during the period of 1980-2010.
  - Suggests China is below its steady state and therefore growing rapidly.

An Increase in the Savings Rate s

- In the steady state, this increases capital per worker and real output per capita.
- In the steady state, there is no effect on the growth rates of aggregate variables.

Figure 6.15
Effect of an Increase in the Savings Rate on the Steady State Quantity of Capital per Worker

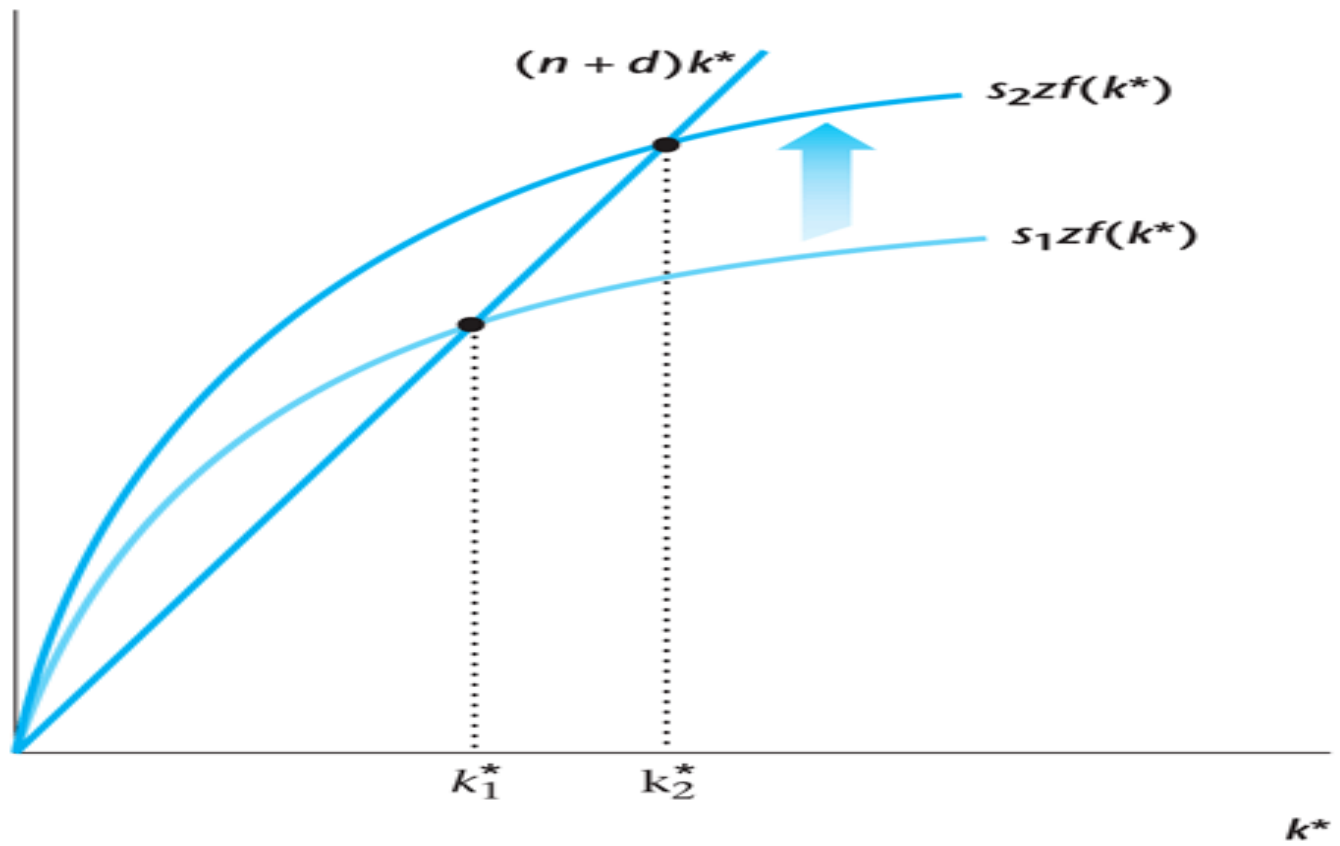


Figure 6.16
Effect of an Increase in the Savings Rate at Time *T*

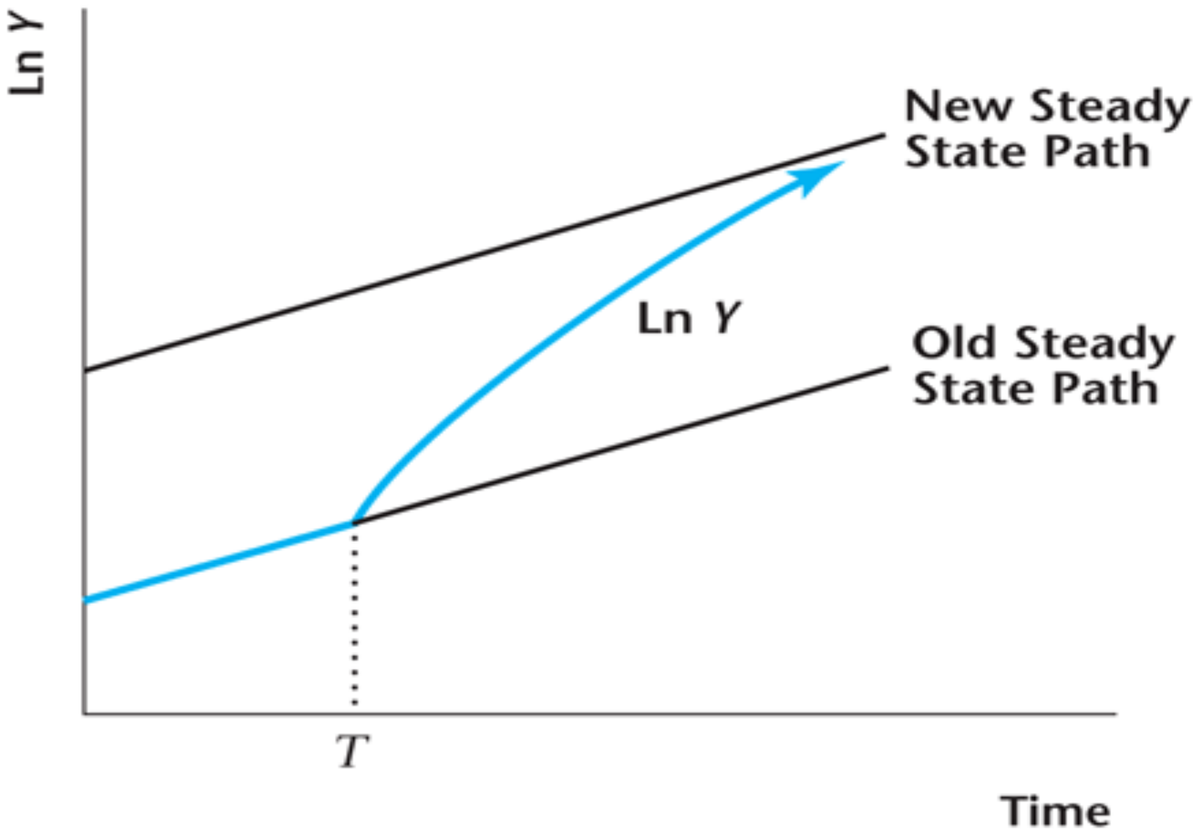


Figure 6.17
Steady State Consumption per Worker

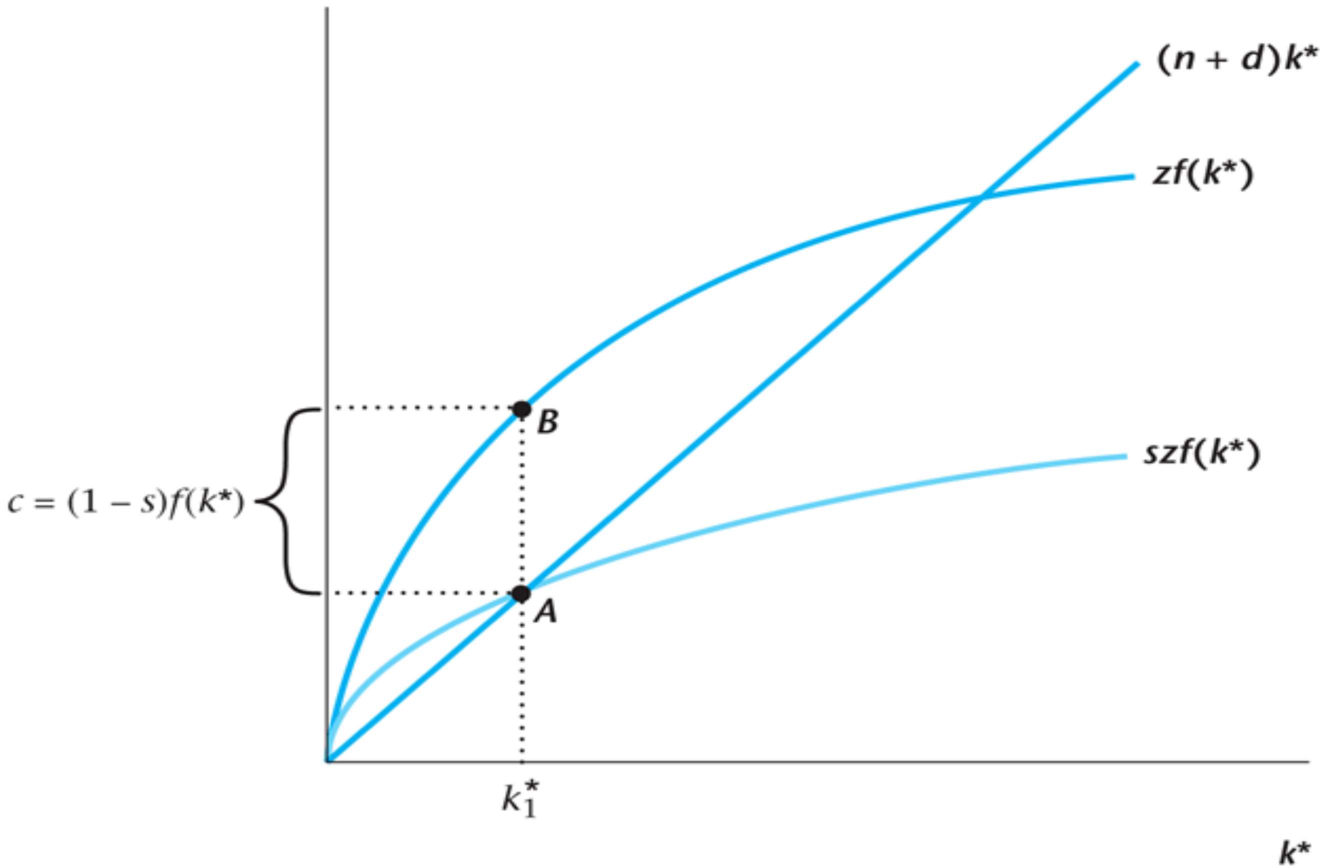
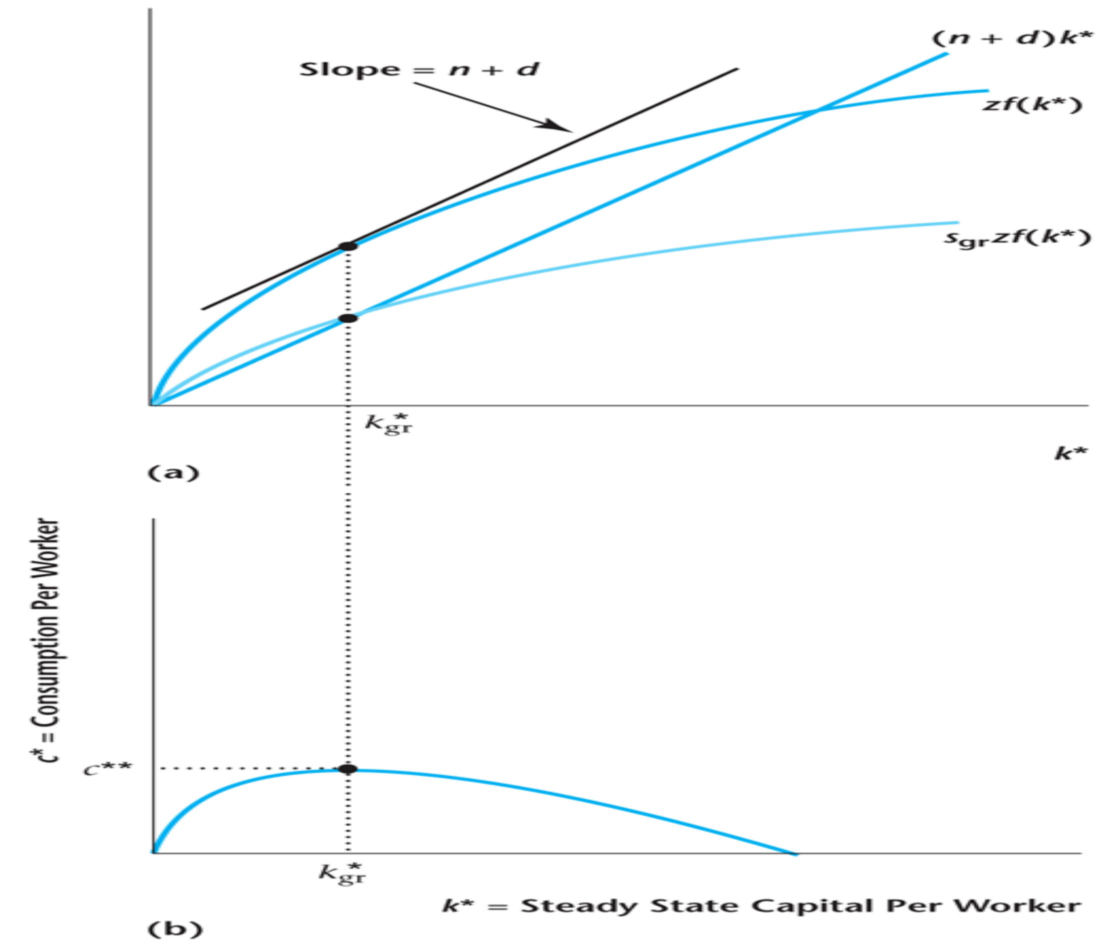


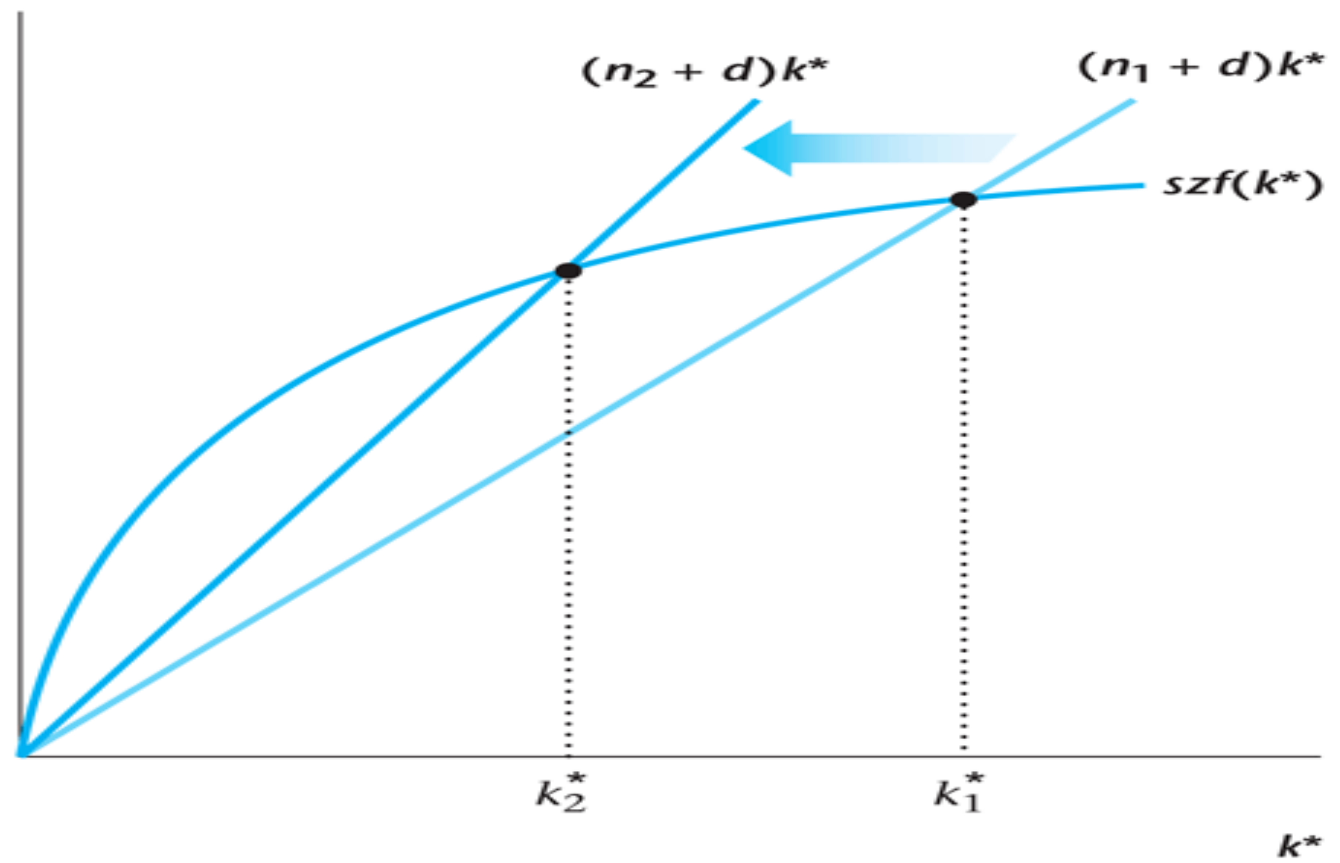
Figure 6.18
The Golden Rule Quantity of Capital per Worker



An Increase in the Population Growth Rate n

- Capital per worker and output per worker decrease.
- There is no effect on the growth rates of aggregate variables.

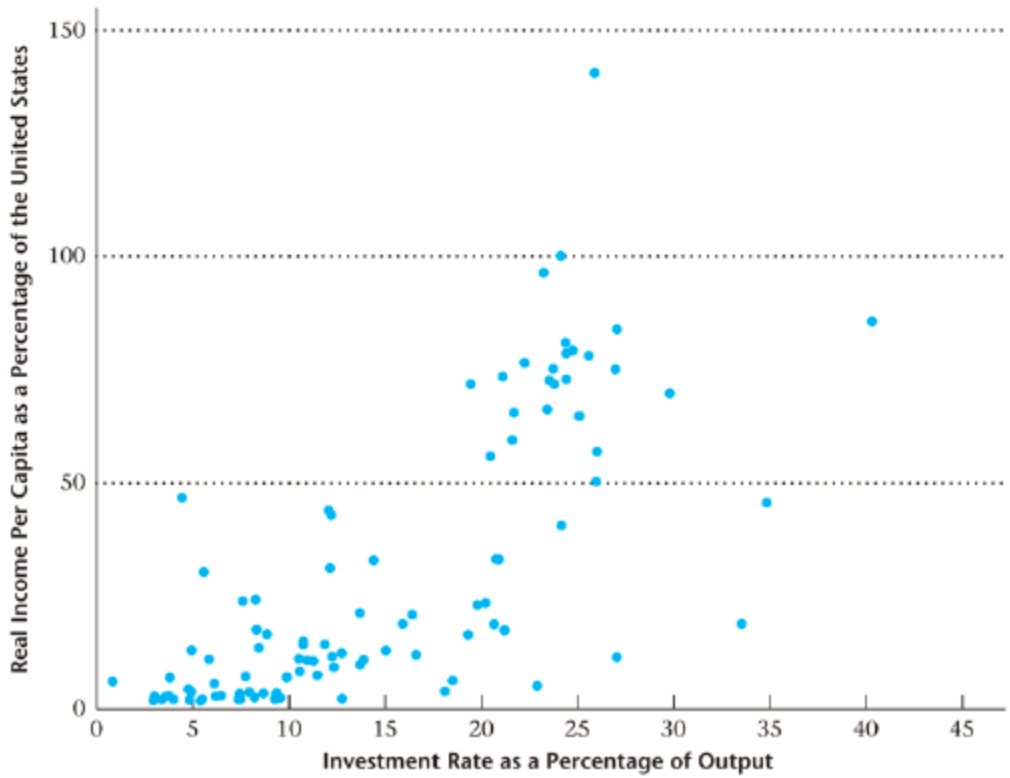
Figure 6.19
Steady State Effects of an Increase in the Labor Force Growth Rate



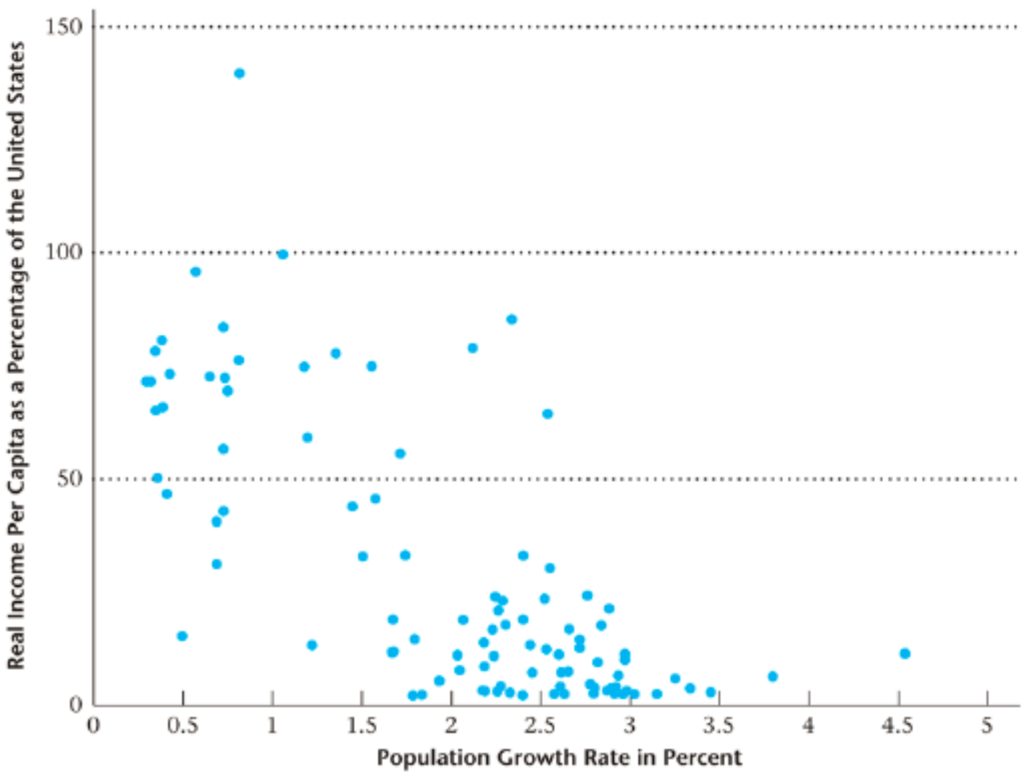
Predictions of The Solow Model

- Richer Countries have a high saving rate.
- Richer Countries have a low population growth rate.

Investment rate and GDP



Population growth rate and GDP



Case Study: S. Korea and Philippines

- In 1960, South Korea and the Philippines were quite similar
    - Per capita GDP: S. Korea=$1800, Philippines=$2200
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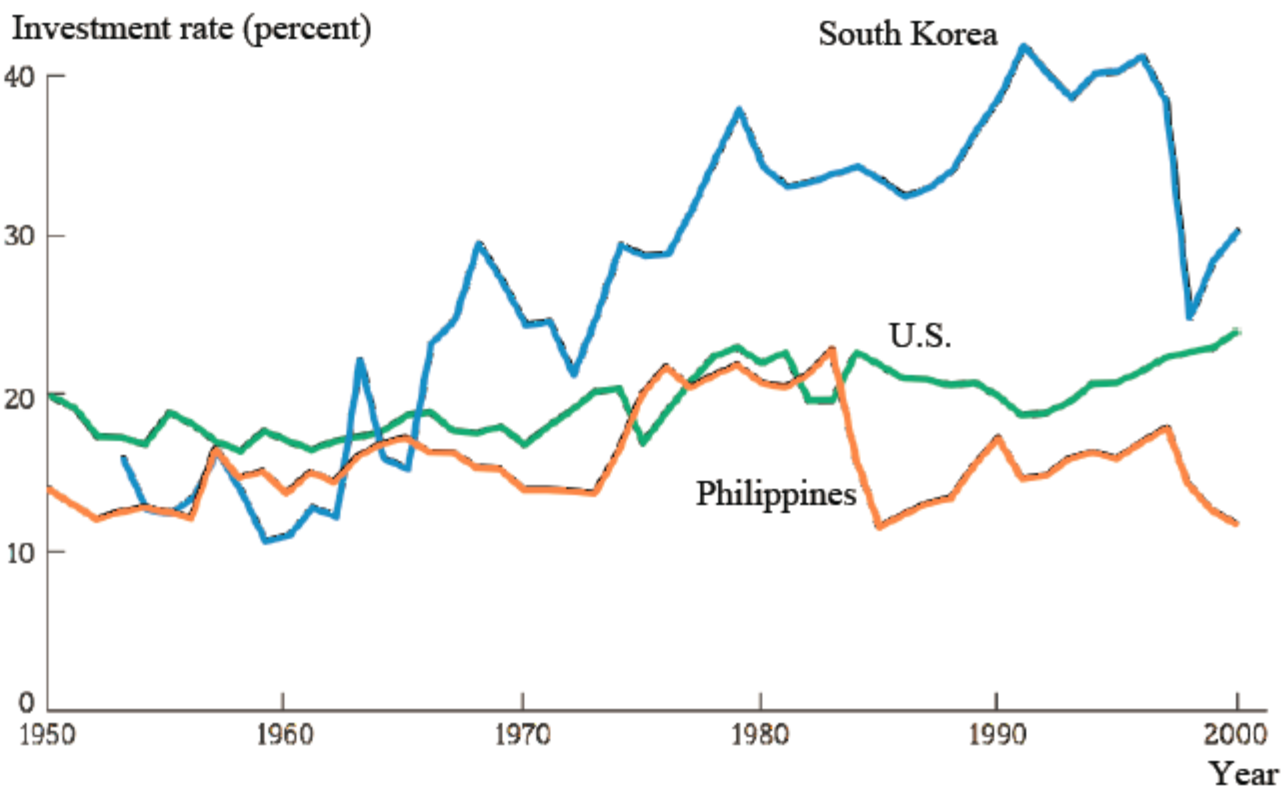
Investment in S. Korean and Philippines



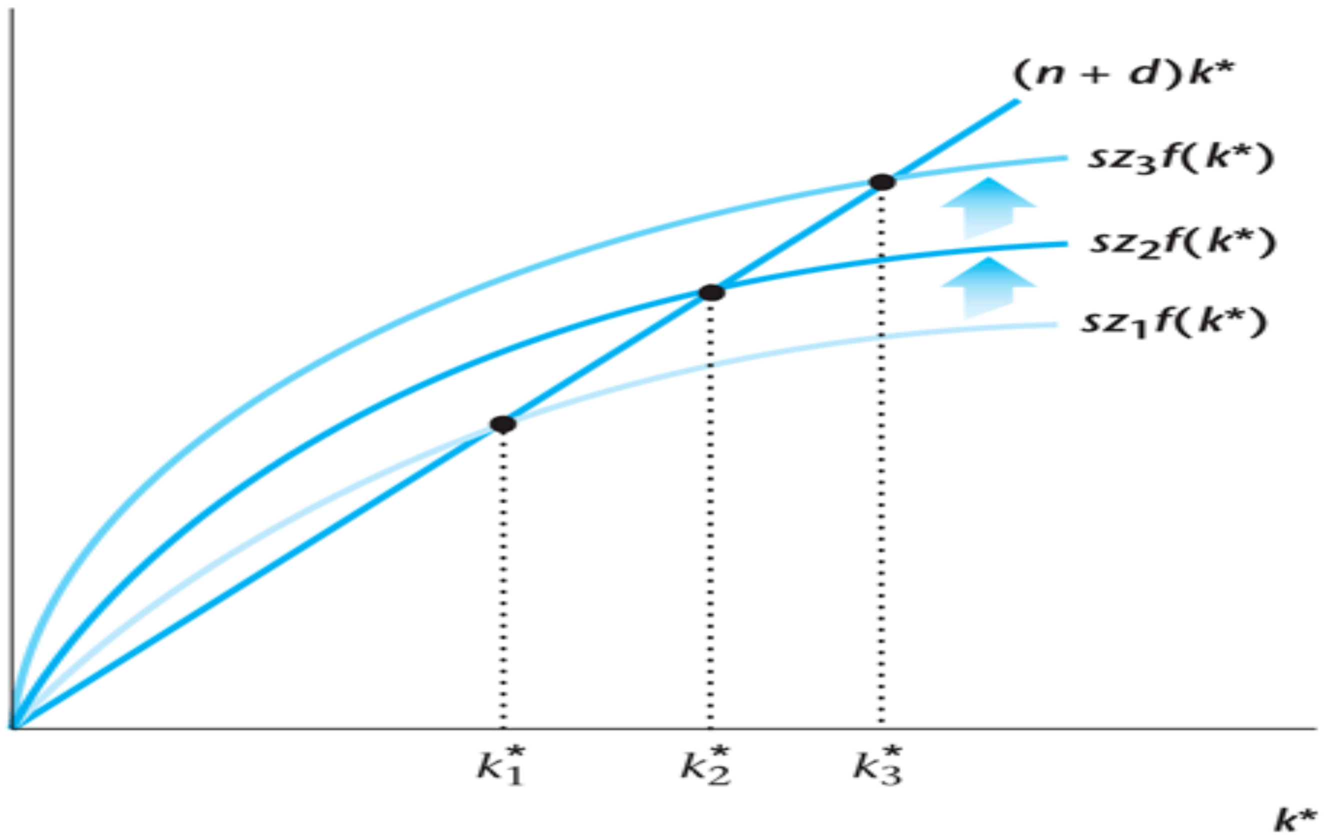
What is the source of Long Term Growth?

- NOT the savings rate: it must always be below *1*.
- NOT decreasing population size: it cannot fall indefinitely.

Increases in Total Factor Productivity z

Sustained increases in z cause sustained increases in per capita income.

Figure 6.20
Increases in Total Factor Productivity in the Solow Growth Model



Strengths and Weaknesses of the Solow Model

- Strengths:
  - it provides a theory that determines how rich a country is in the long run.
  - the principles of transition dynamics allows for an understanding of differences in growth rates across countries.

- Weaknesses:
  - it focuses on investment and capital, while the much more important factor of TFP is still unexplained.
  - It does not explain why different countries have different investment and productivity rates.
  - It does not provide a theory of sustained long-run economic growth.

Convergence

- If two countries are initially rich and poor, but identical in all other respects, they will converge in the long run to the same level and rate of growth of per-capita income.

- Differences in income level across countries explained by differences in *s, n, d, z*.

- Differences in growth rate across countries explained by:
  - Transition dynamics (how far is an economy from its steady state).
  - Differences in the growth rates of *TFP, z*.

Practice Questions

Problem 1
In the Malthusian model, suppose that there is a technological advance that reduces the death rate for each level of standard of living. Using diagrams determine the effects of this on the long-run steady state and explain your results.

Problem 2
Suppose that the depreciation rate decreases. In the Solow growth model, determine the effects of this on the quantity of capital per worker and on output per worker in the steady state. Explain the economic intuition behind your results.

Practice Questions

Problem 3

Consider the Solow model. Suppose an economy begins in steady state and is characterized by the following parameter values: $s=0.2$, $d=0.1$, $z=1$, $N=100$, $n=0$ and with the production function $Y=zK^{1/2}N^{1/2}$. Calculate the growth rate of per capita GDP in the period immediately following each of the changes listed below:

- The saving rate doubles.
- The productivity level rises by 10%.
- An earthquake destroys 75% of the capital stock.

The Solow Growth Model and the 2008-09 Recession

- Attempt to use the Solow growth model to replicate what occurred in the 2008-09 recession in the U.S.

- Model the recession as being caused by a drop in total factor productivity.

- While we know that financial factors played an important role in the recession, it is not outlandish to represent this as a drop in TFP.

Setup of the Recession Experiment in the Solow Model

- Assume reasonable values for the parameters in the Solow model: $d = 0.1$; $n = 0.01$; $s = 0.14$.

- Assume 1 period is a year.

- Assume the production function is Cobb Douglas with capital share parameter 0.36:

$$Y = zK^{0.36}N^{0.64}$$

Experiment in the Calibrated Model

- Temporary (1): TFP falls for one year by 5%, then returns to normal.
- Temporary (2): TFP falls for two years by 5%, then returns to normal.
- Permanent: Permanent decrease of 5% in TFP.

$$Y = zK^{0.36}N^{0.64}$$

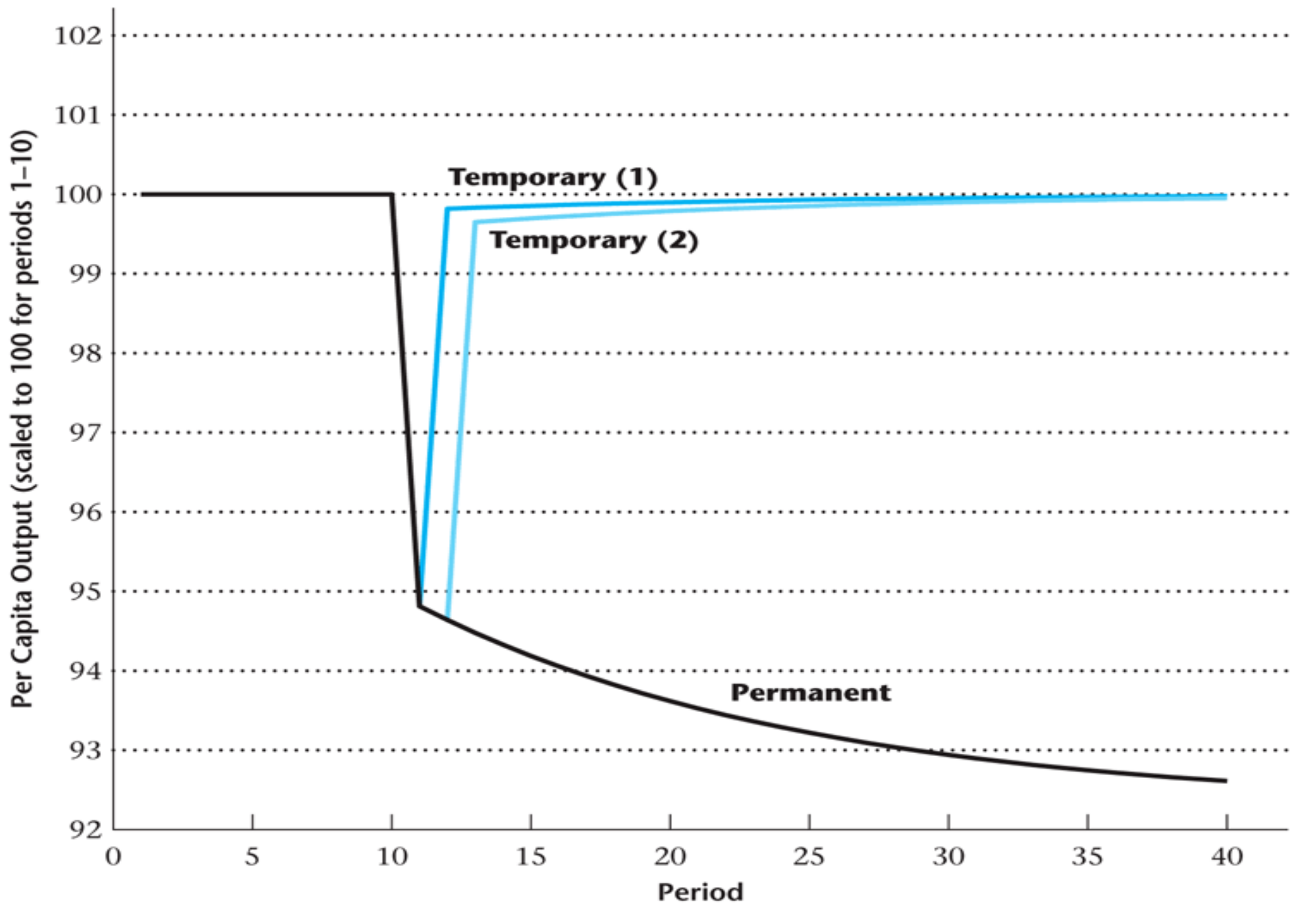
Results

- The longer the drop in TFP persists, real output per capita drops more.

- Why does this occur? The longer the drop in TFP persists, the more investment is foregone, the more the capital stock drops, and the less output there is.

- A persistent shock reduces investment and the capital stock, and this has a lingering effect, even after TFP has returned to normal.

Figure 6.21
Simulation of a Calibrated Solow Growth Model: Temporary and Permanent Declines in TFP



Growth Accounting

An approach that uses the production function and measurements of aggregate inputs and outputs to attribute economic growth to:

(i) growth in factor inputs;

(ii) total factor productivity growth.

Cobb-Douglas Production Function

$$Y = F(K, N) = zK^{\alpha}N^{1-\alpha}$$

Cobb-Douglas Production Function

- A labor share in national income of 64% gives:

$$Y = zK^{0.36}N^{0.64}$$

Solow Residual

The Solow residual is calculated as:

$$z = \frac{Y}{K^{0.36}N^{0.64}}$$

Figure 6.22
Natural Log of the Solow Residual, 1948–2007

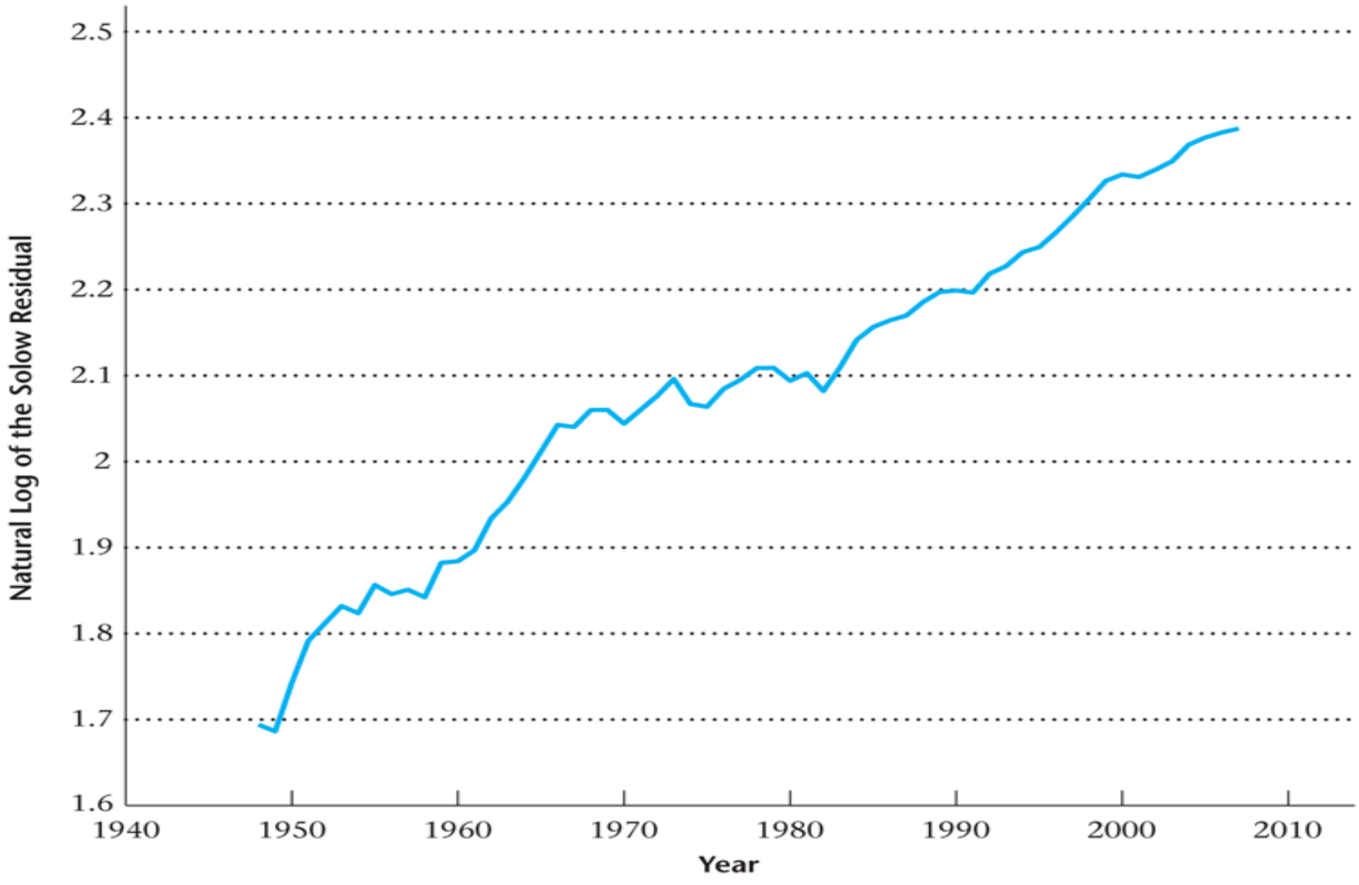
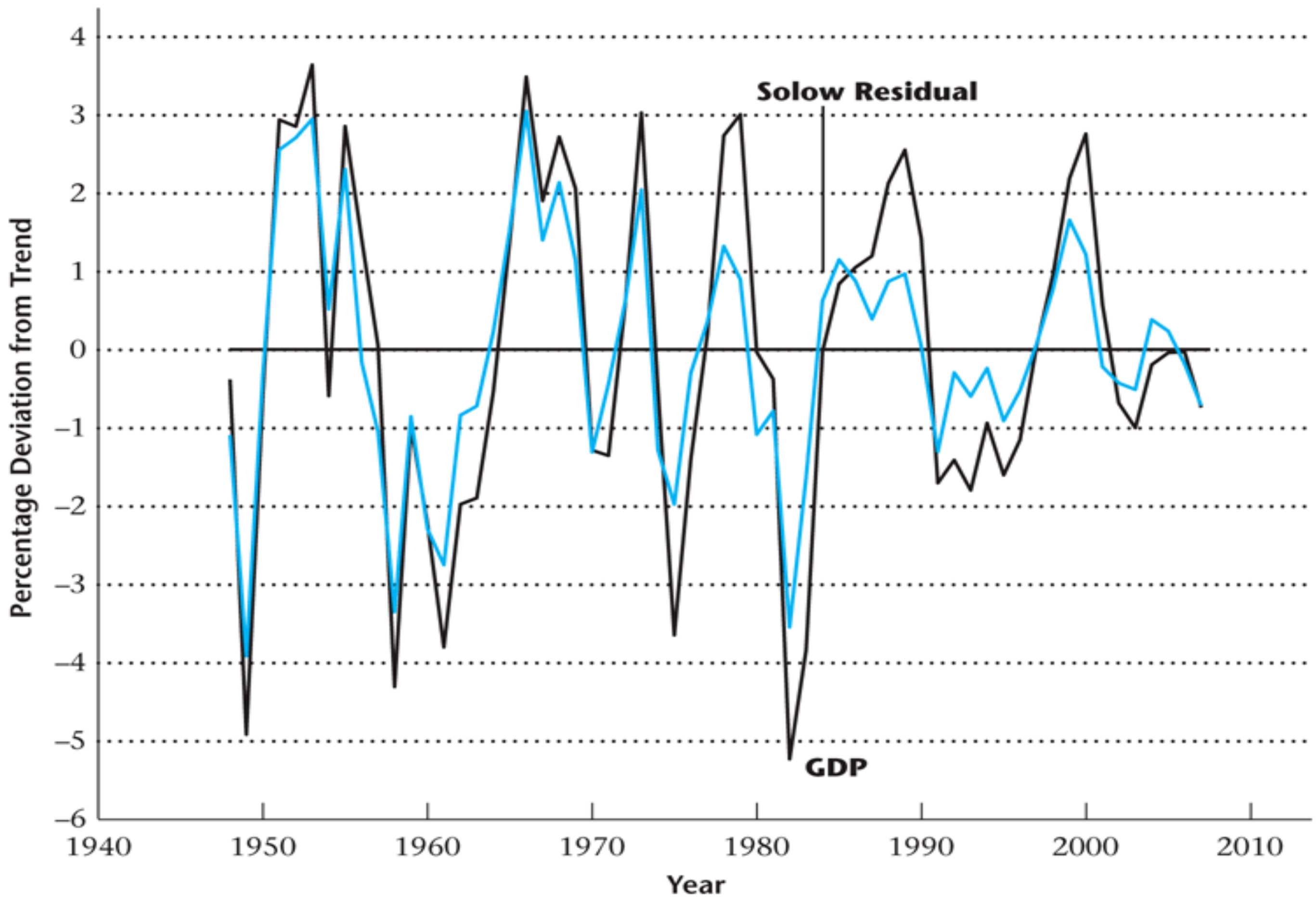


Table 6.1
Average Annual Growth Rates in the Solow Residual

Table 6.1 Average Annual Growth Rates in the Solow Residual

Years	Average Annual Growth Rate
1950–1960	1.42
1960–1970	1.61
1970–1980	0.50
1980–1990	1.05
1990–2000	1.36
2000–2007	0.76

Figure 6.23
Percentage Deviations from Trend in Real GDP and the Solow Residual, 1948–2007



Growth Accounting

- The production function can be written in terms of growth rates:

$$\frac{\Delta Y_t}{Y_t} = \frac{\Delta z_t}{z_t} + 0.36\frac{\Delta K_t}{K_t} + 0.64\frac{\Delta N_t}{N_t}$$

- What is the economic interpretation of this equation?

- The growth rate of output is the sum of TFP growth and inputs growth.

Table 6.2
Measured GDP, Capital Stock, Employment, and Solow Residual

Table 6.2 Measured GDP, Capital Stock, Employment, and Solow Residual

Year	$\hat{Y}$ (billions of 2000 dollars)	$\hat{K}$ (billions of 2000 dollars)	$\hat{N}$ (millions)	$\hat{z}$
1950	1777.3	5991.8	58.89	5.715
1960	2501.8	8602.1	65.78	6.580
1970	3771.9	12557.1	78.69	7.721
1980	5161.7	17273.8	99.30	8.115
1990	7112.5	22877.9	118.80	9.011
2000	9817.0	29917.1	136.90	10.312
2007	11523.9	35910.4	146.05	10.875

Table 6.3
Average Annual Growth Rates

Table 6.3 Average Annual Growth Rates

Years	$\hat{Y}$	$\hat{K}$	$\hat{N}$	$\hat{z}$
1950–1960	3.48	3.68	1.11	1.42
1960–1970	4.19	3.86	1.80	1.61
1970–1980	3.19	3.24	2.36	0.50
1980–1990	3.26	2.85	1.81	1.05
1990–2000	3.28	2.72	1.43	1.36
2000–2007	2.32	2.64	0.93	0.76

Table 6.4
East Asian Growth Miracles (Average Annual Growth Rates)

Table 6.4 East Asian Growth Miracles (Average Annual Growth Rates)

	Output	Capital	Labor	Total Factor Productivity
Hong Kong (1966–1991)	7.3%	7.7%	2.6%	2.3%
Singapore (1966–1990)	8.7%	10.8%	4.5%	0.2%
South Korea (1966–1990)	10.3%	12.9%	5.4%	1.7%
Taiwan (1966–1990)	9.4%	11.8%	4.6%	2.6%
United States (1966–1990)	3.0%	3.2%	2.0%	0.6%